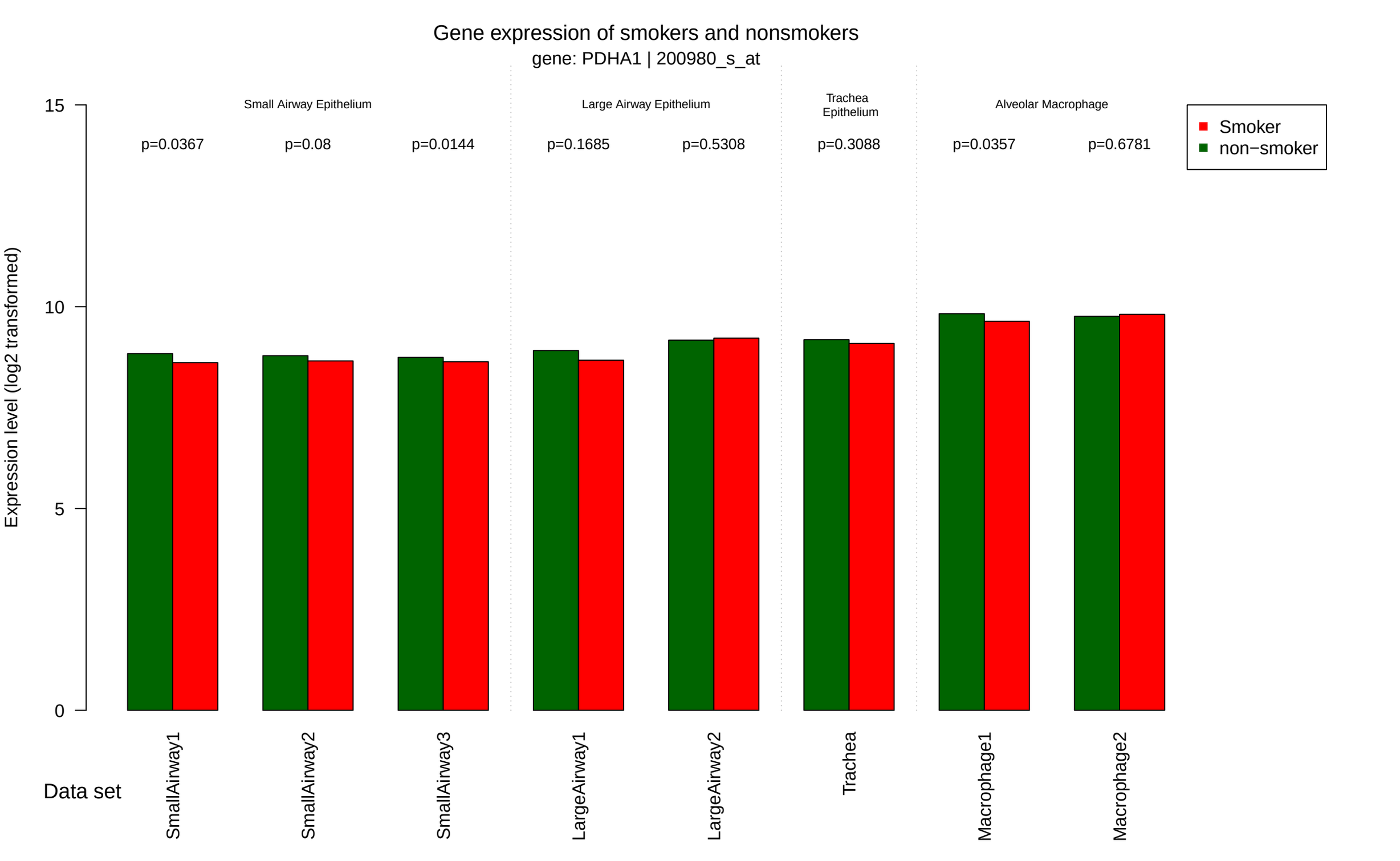
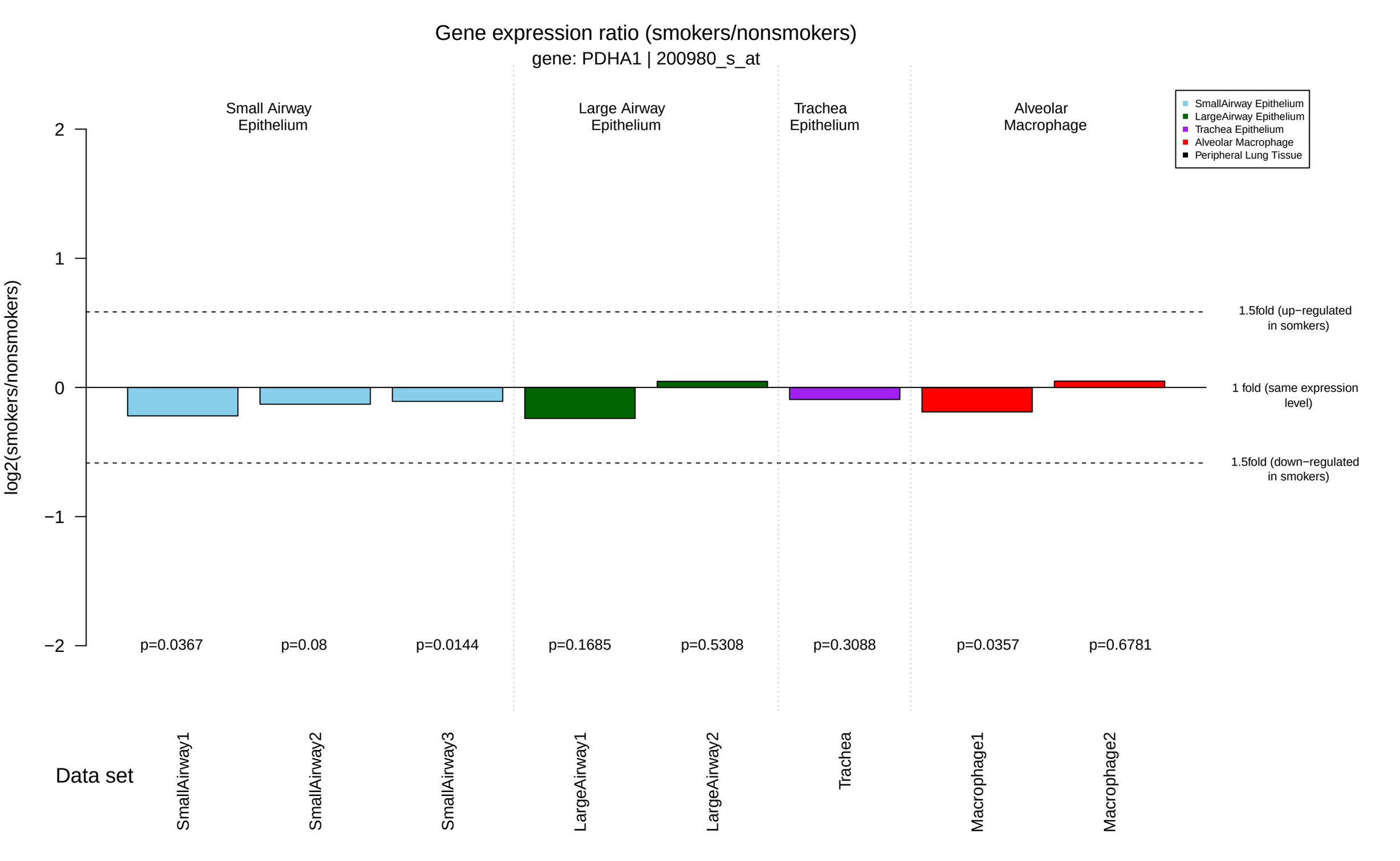
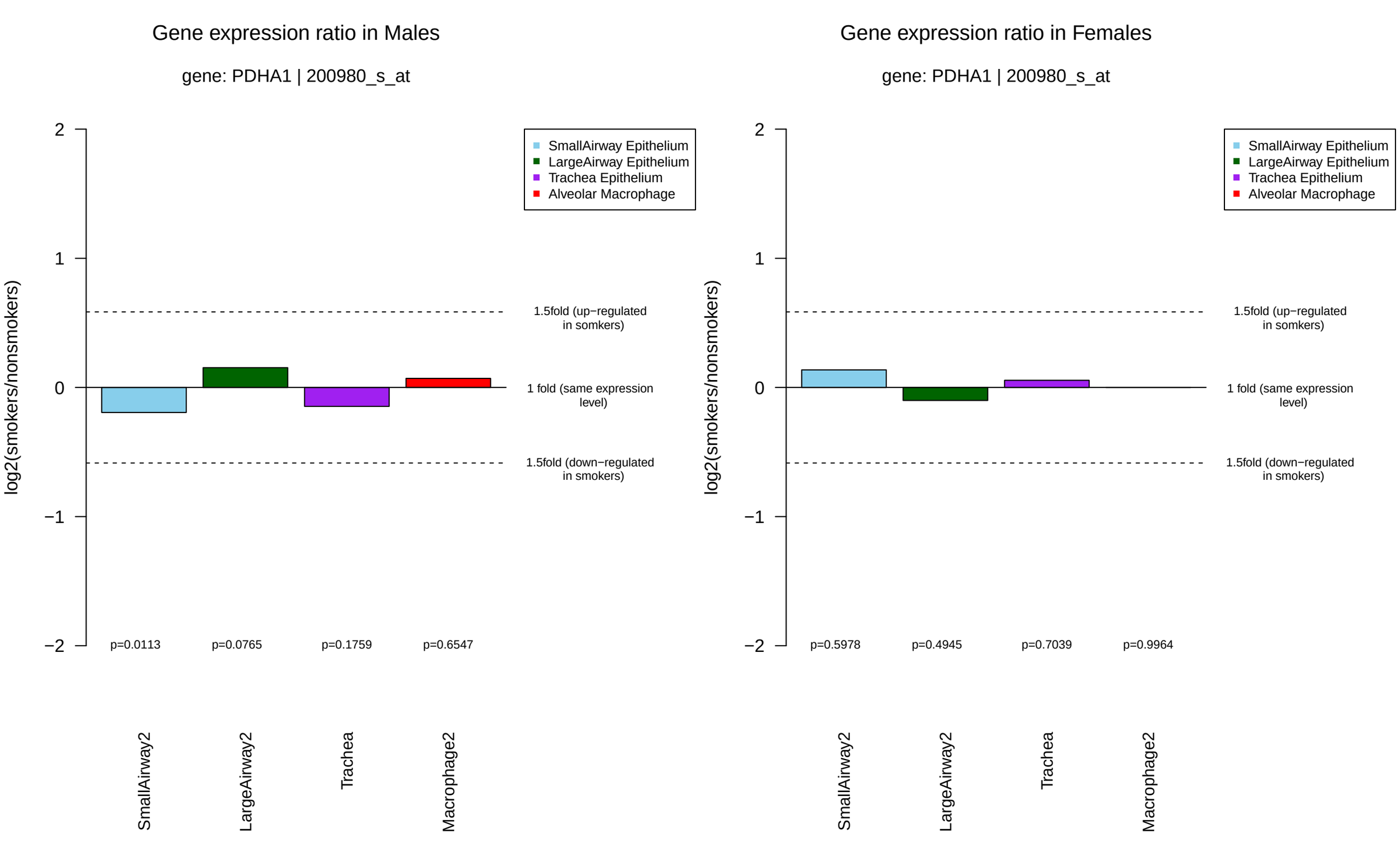
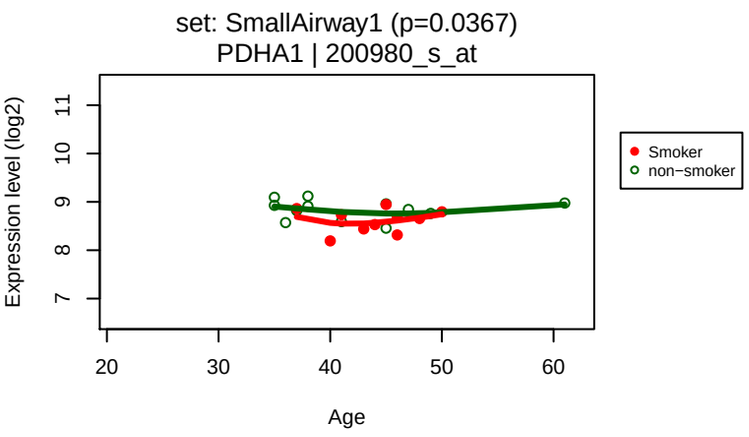
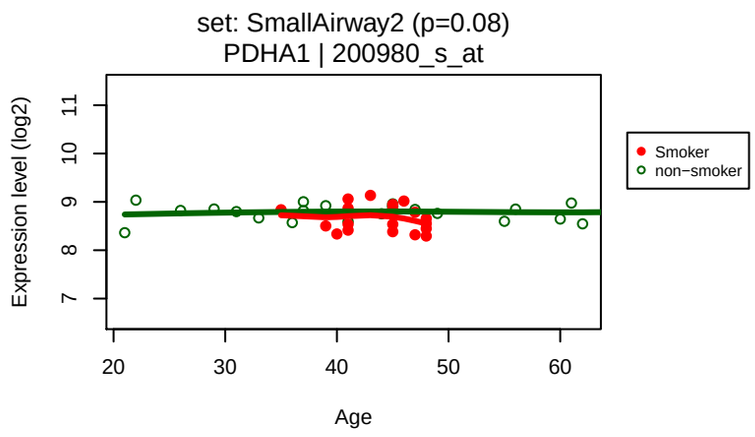
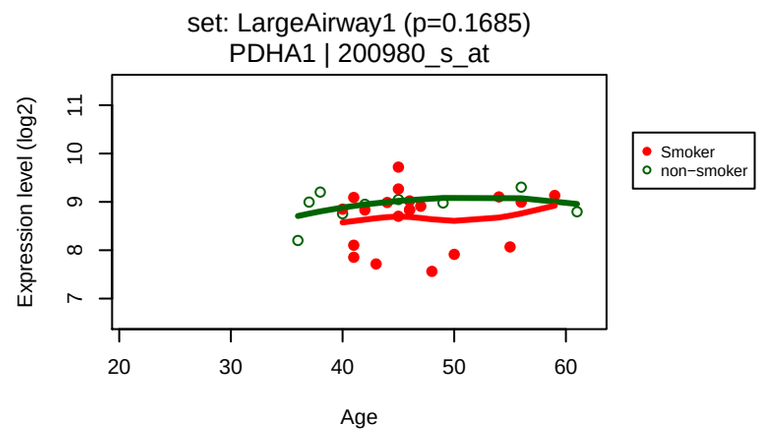

Gene expression of smokers and nonsmokers
gene: PDHA1 | 200980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0367
p=0.08
p=0.0144
p=0.1685
p=0.5308
p=0.3088
p=0.0357
p=0.6781
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

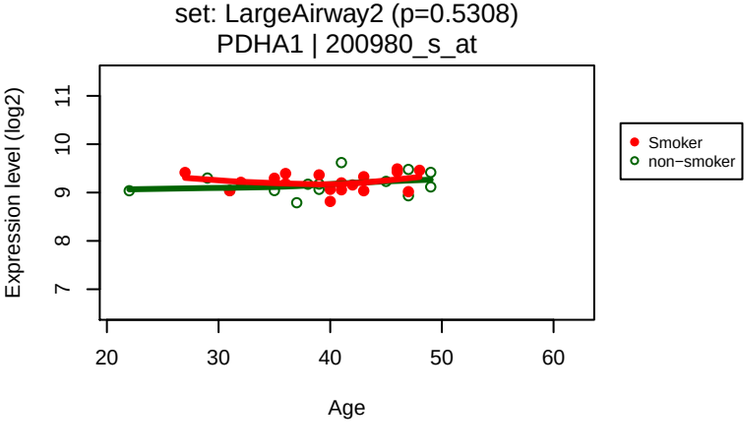

Gene expression ratio (smokers/nonsmokers)
gene: PDHA1 | 200980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0367
p=0.08
p=0.0144
p=0.1685
p=0.5308
p=0.3088
p=0.0357
p=0.6781
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

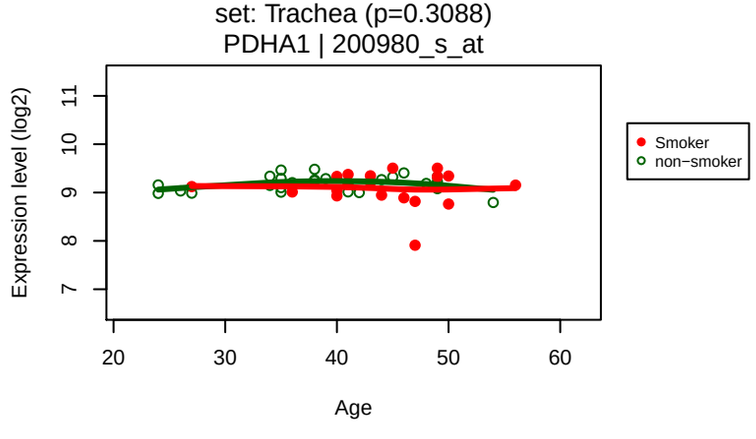

Gene expression ratio in Males
gene: PDHA1 | 200980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0113
p=0.0765
p=0.1759
p=0.6547
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDHA1 | 200980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5978
p=0.4945
p=0.7039
p=0.9964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

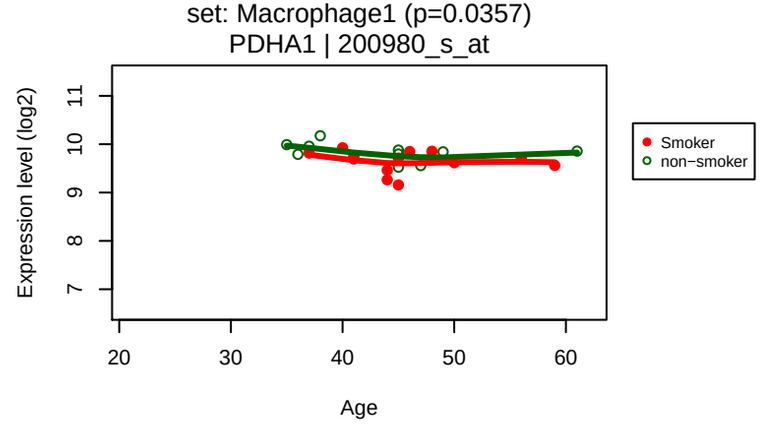

set: SmallAirway1 (p=0.0367)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker

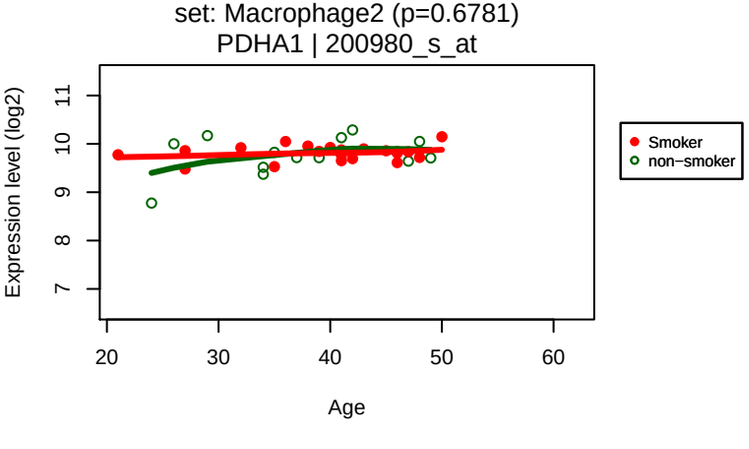

set: SmallAirway2 (p=0.08)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker

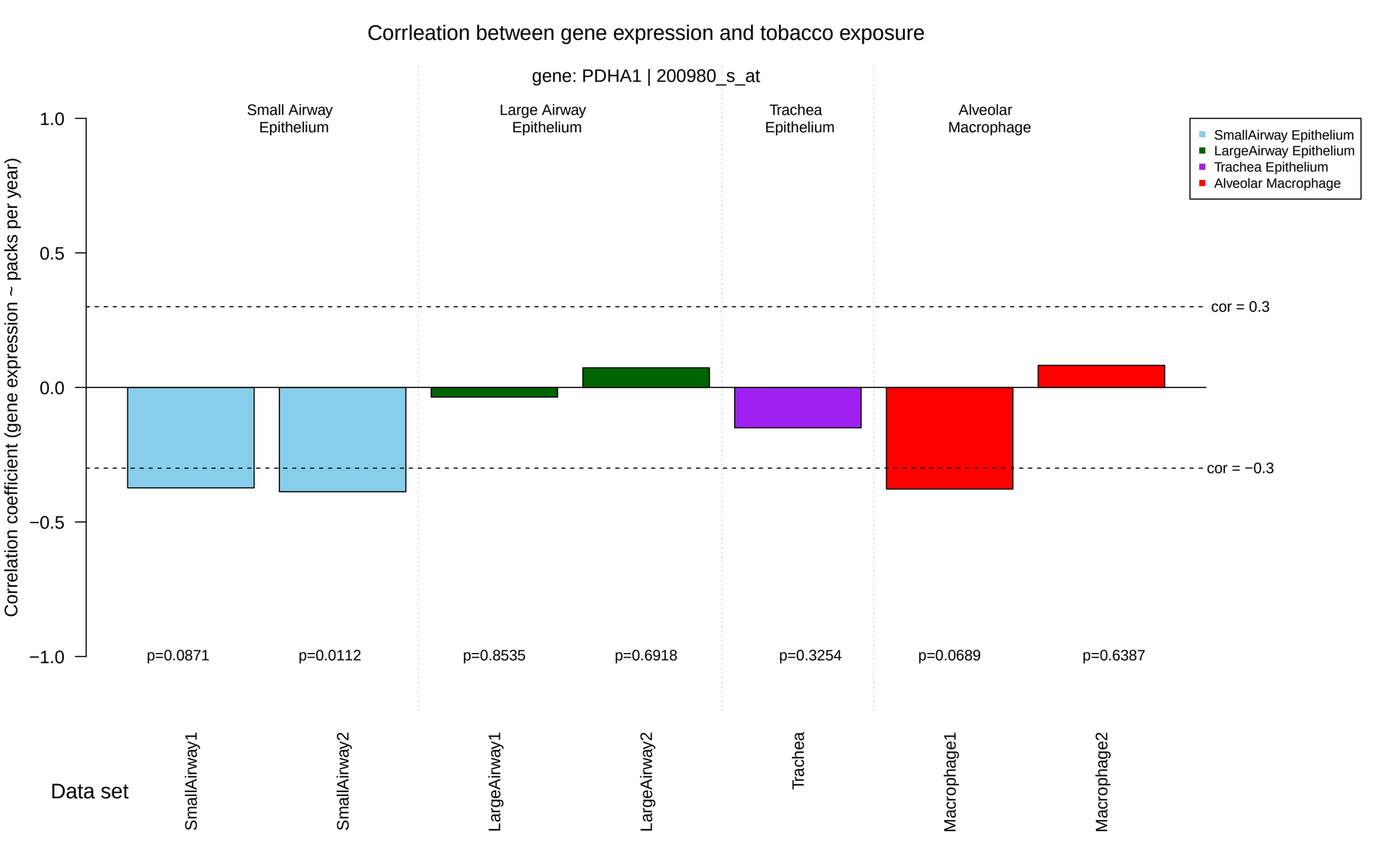

set: LargeAirway1 (p=0.1685)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5308)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3088)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0357)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6781)
PDHA1 | 200980_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDHA1 | 200980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0871
p=0.0112
p=0.8535
p=0.6918
p=0.3254
p=0.0689
p=0.6387
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2